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Video Dancer audit

Read-only audit and design guide. Generated from the editable Markdown sources named inside; change those, then rebuild.

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Video Dancer audit, 2026-10-08 (Opus)

Read-only. vi-dancer `627525f` + working tree, koan-shared `4b46f10`. Threat model: one person on their own PC. Part 1 was redone on 9 October: five agents read the whole app (about 75,000 lines, every file of vi-dancer `src/` and koan-shared `vidancer-ui/src`), then I checked every finding in the code myself. Full write-ups with file, line, path and fix idea: `docs/audit-2026-10-09/` (A saving and files, B money, C back end, D editing screens, E generator screens, plus my verdict on each). This file is the working source for the audit PDF. Nothing here is a fix.

Part 1. Bugs (full re-audit, 9 October)

99 bugs: 18 high, 45 medium, 36 low. 13 of them are unconfirmed: the code path is real, but whether it bites depends on a supplier's behaviour or on timing (marked "unconfirmed"). One agent finding was not real and is left out (an old music-lane gate in export that no project can reach any more).

All 61 existing tests pass. None of them covers the bugs below.

Batch 1 fixed 9 October (committed, not installed): the settings fix plus H1, H3, M10, L2, L5, L6, L7. One rule for every file a save builds on (`fileSave.ts` readIfExists): only a missing file is empty, a lock waits, damage or a lock that never lets go refuses the save and leaves the file as it was. Test: `scripts/test-failed-reads.cjs`.

Batch 2 fixed 9 October (committed, not installed): H4, H5, H6, H8, H9, H10, H11, H12, M14, L11, L12. One mark, `markProviderAccepted()` (jobNotifications.ts), set at every point a supplier hands back a job id or a paid answer; a marked job that fails is never retried, and the window is told `accepted`, so it is never re-queued and its row never says "run it again". ✕ before the submit stops it there (`assertNotStopped()`), Co-Director stop refuses the rest of the reply, Runway THROTTLED is waited for, a Bridge reply that reads like a job was made is never answered with a resend, fal's own submit re-send on a gateway error is stopped, last session's queue is offered once per project per session and each project's rows go to its own file. Tests: test-fal-submit-once.cjs (new), cases added to test-job-notifications, test-bridge-logins, test-bridge-image, test-queue-failure-browser, test-codir-fire-browser.

Batch 3 fixed 9 October (committed, not installed): H7, H13, M4, M12, M17, L8, L9. Video Editor jobs and clip edits now leave a recovery entry the moment a supplier takes them (fal, Higgsfield, Venice; Google's extend has no job id to keep), and recovery lands them as a node or a take the same way the live run does. A Bridge (auto) render sent to Venice is filed as Venice. A local render that finishes after a project switch is filed with its file and added when its project opens. Spend is booked once, in main, in the project that ran the job: live jobs (window and Co-Director plan) in the job wrapper, recovered jobs by recovery; every job carries its estimate (`spend`), and a supplier's own price wins. Enhance holds off the recovery pass while it runs; ✕ in its second stage keeps the first stage's entry; one failed fal image download no longer throws away the other variants. Not booked yet: Co-Director plan Enhance jobs (no estimate in main). Tests: cases added to test-recovery and test-job-notifications.

Batch 4 fixed 9 October (committed, not installed): H2, H15, M29, L3, L4. A result that lands in the background (a render's take, an edit's node, a recovered job, a measured length) is no longer an undo step: what it added is put back into any older snapshot undo or redo restores (`project.ts` landings), so Ctrl+Z never takes a paid result away and redo survives it. Switching timelines is no undo step and undo keeps the timeline in view; the switch clears the Clip Monitor's block and the selection. Undo/redo no longer use a global switch a failed save could leave stuck. Image Gen, Image Editor and Video Editor deletes keep the files (Cleanup reclaims them), so "undoable — Ctrl+Z" is true; the Image Editor ✕ and a single Image Gen take delete now ask. Cleanup keeps a live take's reversed copy. Test: `scripts/test-undo-landings.cjs`.

Batch 5 fixed 9 October (committed, not installed): H14, M5, M6, L1. A duplicate copies the project and its media but never its run state (an approved Co-Director plan, the queue, recovery entries: `project.ts` copyProjectFolder). Sheet build, cross-project clip import and Assembly Concept / Assemble check that the project that asked is still open before they write, and an import commits into the live project (a Ctrl+Z meanwhile no longer leaves clips only on screen). A Co-Director brief carries the project it was typed in; main refuses it anywhere else, and the box resets when the project changes (the last 600 ms of typing before a switch is not saved). Test: `scripts/test-wrong-project.cjs`.

Batch 6 fixed 9 October (committed, not installed): H16, M3, M36, M37, M38, M39, M40, M41, M43, L29, L31. Clip Gen fills its form through one function (clip load, ↩ load Tn's recipe, the empty form after a project switch), so the seed and the links load and restore with the rest and a switch leaves nothing behind. A reload of the same clip (a Co-Director write, an undo) takes only the fields that changed in the store, never the ones being typed; Style Sheet does the same; Effects cancels a pending slider commit only when that block's effects really changed. A pending save goes out on a clip / style / Image Gen switch or panel close instead of being dropped. A new-clip render that lands adopts the typed form (beats included, as does "+ new"). Score length and refinement clamp on blur/Enter; a queued Voice render carries the recipe as pressed; a queued Co-Director message keeps the clip / focus / Image Gen it was typed on; a dropped Watched image joins the Library first; the Sheet Gen name renames on blur/Enter. Not covered: nothing flushes a pending edit when the app window closes. Tests: `scripts/test-forms-browser.cjs` (test:forms-ui), cases added to test-score-browser and test-voice-browser.

Batch 7 fixed 9 October (committed, not installed): M8, M24 to M28, M30, M32 to M35, L19 to L27. Timeline writes answer as soon as the change is in memory (the save finishes in its chain), so quick nudges and gestures no longer snap back; insert-audio is one write. A cut (razor, overwrite, head trim) gives each piece the volume envelope it plays and a fade only on its own edge (`audioEnv.ts` envHead / envTail). A head trim keeps linked sound in sync (main `keepRidersInSync`). Lane audio with a speed is drawn, snapped, keyframed and trimmed as it plays. Timeline keys and Ctrl+C/X/V act only while a timeline or the Timeline Monitor is the active panel. A drop lands in the timeline its panel shows; Bin clips drag as their own type; Assembly defers to the docked Timeline Monitor; reversed blocks hand the forward window to the Video Editor and the Clip Monitor plays the reversed copy. Deletes take linked sound along and respect locked tracks; the Clip Monitor trims like the timeline; a volume band drops its selection on an outside press; the effects baker decodes per block; slot adoption finds its timeline; the Image Editor refs take a multi-drag; a sound-only timeline asks before deleting; a lane drop builds from the current timeline. Tests: `scripts/test-timeline-edits.cjs`, a Ctrl+V case in test-timeline-keys-browser.

High: money, keys, lost work

#	What happens to you	Where	Status
H1	One failed read of a project's spend.json (a scanner or sync lock, a damaged file) and the next paid job saves an empty ledger plus one entry over it. The whole money record of that project is gone. Same mistake as the settings wipe.	`project.ts` readSpend / addSpend	proven (A1)
H2	Deleting an Image Gen job or take, a Video Editor node or an Image Editor branch erases the files at once (not to the Recycle Bin). The dialogs say "undoable, Ctrl+Z", but undo only brings back entries pointing at missing files. The Image Editor ✕ deletes a whole branch with no question. Paid edits are lost.	`project.ts` deleteImageGen / deleteImageGenTake / deleteVideoEditBranch / deleteEditBranch	proven (A2, D12, E21)
H3	One failed read of the Co-Director's learned rules or of your edited prompting guide, and the next "remember this" or guide edit writes over the file: all earlier rules gone, or your guide replaced by the built-in one.	`guides.ts` readLearned / readGuideText	proven (A10)
H4	A "render now" job the supplier already accepted gets sent again when its status checks hit a busy reply (429 / 503) six times in a row: the app reads it as "rate limited, moved to queue". You pay twice. Queue rows that fail this way say "Run it again", which does the same by hand.	`store.ts` busy re-queue, every provider's poll loop	proven (B1)
H5	Runway's THROTTLED (a job accepted and waiting its turn) is treated as failed: "render now" rows are sent again, recovery throws the job away, Runway image edits are lost.	`runwayMcp.ts`, `bridgeImage.ts`	proven in code; Runway's docs call THROTTLED a waiting state (B2)
H6	Bridge: a later error that mentions credit, quota or payment moves the job to your next login or supplier even after the first one took it.	`bridge/index.ts` onBestLogin, `bridgeAuto.ts`	path proven; a double charge depends on the supplier (B3, was 8 Oct #2)
H7	Video Editor jobs (v2v, edits, lip-sync) leave no recovery record. A quit, crash or ✕ while one runs loses the paid video. Venice's cancel message promises a recovery that cannot happen.	`index.ts` runV2vEngine	proven (B5, was 8 Oct #5)

#	What happens to you	Where	Status
H8	A Co-Director plan automatically retries a paid image edit or video edit that failed after the supplier charged (a failed download, a project switch). You pay again.	`jobNotifications.ts` finish / retryFailed	proven (B6)
H9	"N queued renders from last session, click to re-queue" also offers rows that are still in the live queue, or that you cancelled. One click runs them twice.	`store.ts` offerQueueRestore	proven (B8, was 8 Oct #4, wider)
H10	Pressing stop on the Co-Director does not stop the other renders already asked for in that reply. They still queue and bill.	`codir.ts` tool loop	proven (B9)
H11	A Bridge reply the app can't read makes it send the same paid generate call again, up to 3 to 5 times.	Bridge adapters (Higgsfield, Runway, Magnific, Pika, image lane)	unconfirmed; one paid job was already misreported this way (B15)
H12	✕ pressed while a running job is still uploading its refs does not stop it: the job is sent and billed after you cancelled.	`fal.ts`, `venice.ts`, `astria.ts`, `modelark.ts` cancel	proven (B19)
H13	A "Seedance (Bridge)" render that the router sends to Venice is recorded for recovery as a Higgsfield job. If anything interrupts it, the paid Venice video is never collected.	`index.ts` resumePendingRenders, `bridgeAuto.ts`	proven (B21)
H14	Duplicating a project copies its approved Co-Director plan. Opening the copy makes the Co-Director carry on there with no GO and spend the plan's remaining allowance a second time.	`index.ts` project:duplicateAt	proven (C12)
H15	Ctrl+Z after a render lands removes the paid take (every landing is an undo step). The next edit wipes redo; the file is then orphaned and "delete unused media" erases it.	`project.ts` saveProject history + addTake	proven (D1)
H16	Clip Gen: a "render now (new clip)" that lands selects the new clip and reloads the form, wiping everything you typed while it rendered.	`RenderPanel.tsx` completion watcher	proven (E1)
H17	Seedance on ModelArk 2.5, Venice and the Bridge lanes: when refs carry the default @imageN names and aren't attached in order, several mentions end up pointing at one picture. A paid render with the wrong characters, and the preview looks right.	`compile.ts` applyRefTags	proven, logic run on the code (E12)
H18	Every installer ships the private koan-shared source (343 files) and the 4 draft MOODS.	build config	proven on the installed 0.26.0 (was 8 Oct #3)

Medium: a workflow breaks or does the wrong thing

#	What happens to you	Where	Status
M1	The Queue shows nothing while Image Gen renders: Image Gen runs on its own line inside its panel and never enters the Queue.	`ImageGen.tsx` line / drain	proven, your report 9 Oct (F1)
M2	Deleting a Library music track gives no warning and leaves its timeline blocks silent ("used by" still counts the old music lane).	`Library.tsx`, `project.ts` deleteAudio	proven (A3, D11)
M3	Clip Gen, Style Sheet and the Image Gen prompt drop your last edit if you switch item or project within half a second. Every Co-Director write (and some landings) reloads Clip Gen and the Style Sheet over what you're typing; the Effects panel drops a slider move on the same signal.	`RenderPanel.tsx`, `StyleSheet.tsx`, `ImageGen.tsx`, `EffectsPanel.tsx`	proven (A4, E7)
M4	A local LTX / H3 render that finishes after a project switch is never added, though the message says it will be next time you open that project.	`index.ts` render:run local branch	proven (A6)
M5	Sheet build and cross-project clip import write into whichever project is open when they finish; Ctrl+Z during an import leaves clips that exist only on screen.	`project.ts` buildSheetFromTake, importClipsFrom	proven (A7)
M6	Assembly Concept / Assemble that finish after a project switch overwrite the other project's outline or give it a timeline of clips it doesn't have.	`assembly.ts`	proven (A8)
M7	The Co-Director's save_mood and dropping a .vmd overwrite an installed MOOD of the same name without asking.	`moods.ts` saveMood, importMoodPaths	proven (A9)

#	What happens to you	Where	Status
M8	Quick timeline edits are thrown away: five fast "." nudges move a block one frame; a second gesture right after the first snaps back.	`project.ts` stale-rev check, `store.ts`	proven (A13)
M9	Once saves are failing (project folder renamed in Explorer, disk full), closing the window leaves a hidden process. Video Dancer won't open again until it's killed in Task Manager; Open and New do nothing.	`index.ts` before-quit	proven (A14)
M10	With an unreadable settings.json, Open and New Project switch in the back end, then stop: the window never shows the project. Caused by the uncommitted settings fix.	`index.ts` noteRecent	proven (A12, C7)
M11	Video Editor lists Bridge and local engines for "re-gen with refs", then sends them to fal (uploads to fal, then fails).	`VideoEditor.tsx`, `index.ts` runV2vEngine	proven (B4)
M12	The spend ledger books a render that finishes after a switch to the open project, and never books Co-Director plan jobs or recovered renders.	`store.ts`, `project.ts` addSpend	proven (B7, was 8 Oct #8, wider)
M13	Bridge upload caches are per file, not per login: a ref uploaded on login 1 is reused on login 2, which can't see it.	Bridge adapters	unconfirmed (B11)
M14	A queue save during a project switch can write project A's queued renders into project B.	`index.ts` queue:save	unconfirmed, timing (B12)
M15	Bridge adapters take any video or image link in a reply as the result, sometimes before checking for failure.	`higgsfieldMcp.ts`, `bridgeImage.ts`, `pikaMcp.ts`	unconfirmed (B13)
M16	Two jobs on the same cut window a few seconds apart: the second uploads the first's half-written cut (and the Video Editor caches that broken upload).	`index.ts` resolveRefMedia, vedit:run	proven code path, timing (B20, C5)
M17	A running fal Enhance gets "recovered" by the 30-second recovery loop: an extra half-done copy lands. Cleanup doesn't wait for Enhance either.	`index.ts` enhance:run	proven (C1)
M18	✕ does nothing on a local LTX / H3 render waiting for the GPU or starting up; local scores (ACE, MiniMax) can't be cancelled at all.	`ltxLocal.ts`, `audioLocal.ts`	proven (C2)
M19	The MiniMax Music worker's output is never read, so a chatty first run can freeze it and every local job behind it.	`runtime.ts` script workers	unconfirmed (C3)
M20	Every job has a hidden 2-hour deadline from the click (including time waiting for the GPU); at 2 h the app cancels it.	`appJobs.ts`, `jobNotifications.ts`	proven (C8)
M21	Adding or forgetting a LoRA folder, a first Civitai LoRA download, deleting a checkpoint or a local model kills the engine mid-render.	`index.ts` addLoraFolder and others	proven (C13)
M22	Re-picking a display for the output mirror orphans the new mirror: Esc and "off" stop working and the app won't quit when the main window closes.	`index.ts` openMirror	proven (C16)
M23	Krea 2 doesn't tell the shared engine which model it loads, so LTX and Krea 2 models can pile up in one warm engine.	`krea2Local.ts`	unconfirmed (C17)
M24	The razor keeps the whole volume envelope and both fades on both halves: a dip at the cut, keyframes shifted on the right half.	`project.ts` sliceTimelineItem	proven (D2)
M25	Timeline keys fire from other panels once the timeline was clicked: I / O in Clip Gen also set the timeline's in / out; C, Q, W, M, 1-9 act on the timeline from the Bin.	`Timeline.tsx` key handler	proven (D3)
M26	A clip dropped on the main Timeline lands in another timeline when a timeline window was clicked last.	`Timeline.tsx` drop	proven (D4)
M27	"Insert audio → A n" never mutes the block (its second save is refused), so the sound plays twice.	`Timeline.tsx`	proven (D5)
M28	Lane audio with a speed is drawn, snapped, keyframed and trimmed as if at normal speed.	`Timeline.tsx`	proven (D7)
M29	Switching timelines is saved as an undo step and wipes redo; the Clip Monitor can jump to a different block in the new timeline.	`store.ts` selectTimeline, `project.ts` setActiveTimeline	proven (D8)
M30	Any text drag onto a video track (an Effects reorder handle, browser text) makes a phantom 5 s block that cuts out the footage under it.	`dragDrop.ts` (clip MIME is text/plain)	proven (D10)

#	What happens to you	Where	Status
M31	Video Editor keeps the previous source's selected node, so Generate on the next source fails ("the source video is gone").	`VideoEditor.tsx`	proven (D13)
M32	With Assembly open next to the Timeline Monitor, sound plays twice and gaps play at double speed (Assembly mounts a second Monitor).	`Assembly.tsx`	proven (D14)
M33	After any timeline Ctrl+C, Ctrl+V in the Library pastes that block onto the timeline over footage instead of importing the image.	`App.tsx`, `tlClipboardKeys.ts`	proven (D17)
M34	Linked lane audio drifts out of sync when its block's start is trimmed.	`project.ts` syncLinkedAudio, `Timeline.tsx`	proven (D20)
M35	On a reversed block, the Clip Monitor and "edit this take v2v" / lip-sync use the wrong part of the take; the paid v2v job runs on that wrong section.	`SourceMonitor.tsx`, `Timeline.tsx` editVideo	proven (D22)
M36	Clip Gen seed isn't saved per clip or restored by "load recipe": a re-render doesn't reproduce the look.	`RenderPanel.tsx`	proven (E2)
M37	"Load Tn's recipe" keeps today's prompt link, field links, overrides and linked characters, so the re-render sends a different prompt.	`RenderPanel.tsx` restoreRecipe	proven (E3)
M38	Score length clamps on every keystroke: typing 30 gives 50 (ACE: 100). Same for Beatoven refinement.	`ScoreCard.tsx`	proven (E8)
M39	A queued Voice render speaks whatever the card holds when it starts, not what was there when you pressed + Queue.	`VoicesCard.tsx` payload	proven (E9)
M40	A message typed while the Co-Director is busy is sent with the clip that's selected when the run ends, not the one you meant.	`store.ts` codir queue	proven (E10)
M41	A Watched image dropped on a Clip Gen slot is never saved to the clip and has no @name.	`RenderPanel.tsx` buildPromptData	proven (E11)
M42	Sheet Gen renders are booked at $0; its Nano Banana lane skips the Queue and the ledger.	`SheetGen.tsx`	proven (E14)
M43	Clip Gen "+ new" saves the form as a clip but drops the beats.	`RenderPanel.tsx`	proven (E15)
M44	Watched "→ send both" puts only one LoRA into Image Gen, at weight 1.	`store.ts` imageGenUseSeed	proven (E17)
M45	Civitai Get + Use: a different version with the same filename counts as already downloaded.	`index.ts` imagegen:civitDownload	proven (was 8 Oct #7)

Low: rare, small, or cosmetic in behaviour

#	What happens to you	Status
L1	A Co-Director brief typed just before a project switch is written into the new project's brief. (A5)	proven
L2	A settings read that fails for over 4 s can still rebuild the recents list, Open-window folders or workspaces from empty. (A11)	proven
L3	A failed save during Ctrl+Z switches undo history off for the rest of the session. (A15)	proven
L4	Cleanup deletes the reversed copies of live takes; reversed blocks play black until the timeline changes. (A16, C10)	proven
L5	The Photoshop watch list, the render-failure history and queue.json are rebuilt from empty on a failed read. (A17)	proven
L6	A damaged codir-jobs.json blocks every render and export in that project, with no way out in the app. (A18)	proven
L7	A settings read that fails at launch moves the models folder to the default for the session. Likely no effect on your PC. (A19)	unconfirmed
L8	Image Edit on fal: one failed download throws away every paid variant. (B10, was 8 Oct #6)	proven
L9	Cancelling a two-stage fal Enhance in stage 2 drops the recovery of the finished stage 1. (B14)	proven
L10	ElevenLabs / Google score and voice: ✕ or the timeout drops a job the supplier may still bill. (B16)	unconfirmed
L11	A Co-Director fire that times out (8 s) stays armed and can render later. (B17, E5)	unconfirmed
L12	The fal client retries a submit on gateway errors, which may create a second paid request. (B18)	unconfirmed

#	What happens to you	Status
L13	A second launch while the app runs may kill the first one's local engines. (C4)	unconfirmed
L14	Export, fx passes, probes, cuts and Sheet Gen spawn ffmpeg / python without hiding the console window. Do you see black windows when you export? (C9)	proven in code, open
L15	Export and a Co-Director timeline watch at once: export ✕ stops the watch instead, and the export's end deletes the watch's bakes. (C11)	proven
L16	Mixed-fps timelines with fps on auto: effect spans can bake at one rate and export at another. (C14)	proven
L17	Gemini watch retries a request that timed out, so one slow watch may bill up to three times. (C15)	unconfirmed
L18	Reference copies in the engine's scratch folder are never deleted: 869 files, 2.4 GB on your PC today. (C18)	proven
L19	✕, "remove (leave gap)" and "ripple delete" leave the block's linked lane audio behind. (D6)	proven
L20	Clip Monitor in / out handles extend a slot over the next block. (D9)	proven
L21	A volume keyframe selection left in an expanded lane takes over Delete. (D15)	proven
L22	Effects export: the same take on two tracks shares one decoder, so the offset copy bakes the wrong frame. (D16)	proven
L23	A locked track doesn't stop deletes. (D18)	proven
L24	A render started from a timeline slot isn't adopted if another timeline is active when it lands. (D19)	proven
L25	Dragging several Library images onto the Image Editor's refs adds only the one grabbed. (D21)	proven
L26	Deleting a timeline with only sound on it skips the confirm. (D23)	proven
L27	Moving a lane audio block writes back the video items from when the drag began. (D24)	proven
L28	LoRA side panel: a download refused up front sticks at 0% forever. (E4)	proven
L29	Sheet Gen name box can't take a space or be cleared. (E6)	proven
L30	Clip Gen t2v: parked refs still rewrite the prompt's @names on the wire. (E13)	proven
L31	Clip Gen project switch leaves refs, links, LoRAs and dials from the old project. (E16)	proven
L32	Voice card: switching engine keeps the other supplier's voice id, which then fails. (E18)	proven
L33	Opening Enhance from Optical Flow overwrites your remembered Enhance settings. (E19)	proven
L34	LoRA side panel and Models card read "has a Civitai key" once; a key saved after still blocks downloads. (E20)	proven
L35	Clip Gen: LTX LoRAs ride into an H3 render after a model switch. (E22)	proven
L36	LoRA switch tooltip says off = none, but unleashed still runs. Your call: tooltip or switch. (was 8 Oct #9)	proven

Still open from 8 October: the Civitai key on a cross-site redirect (needs a local redirect test). The 8 October lows not listed above were re-checked by the agents and stand as written there (edge cases, no money or lost work).

How the fixes group

Most of these share a few root causes. Fixing the cause once fixes every bug in its group.

Batch	The one cause	Bugs
1. Failed read = empty, then saved over (FIXED 9 Oct)	Any read error is treated like "no file yet"	H1, H3, M10, L2, L5, L6, L7, plus committing the settings fix
2. An accepted job is never sent twice (FIXED 9 Oct)	Nothing marks a job "the supplier has it" in one place	H4, H5, H6, H8, H9, H10, H11, H12, M14, L11, L12
3. Every paid job can be recovered and booked (FIXED 9 Oct)	Recovery and spend are wired per lane, and some lanes were skipped	H7, H13, M4, M12, M17, L8, L9
4. Deletes and undo (FIXED 9 Oct)	Files deleted outright; background writes and timeline switches recorded as undo steps	H2, H15, M29, L3, L4
5. Results land in the project that asked (FIXED 9 Oct)	Some long jobs write to whatever project is open	H14, M5, M6, L1

Batch	The one cause	Bugs
6. Forms keep what you typed (FIXED 9 Oct)	Autosave timers cleared instead of flushed; reloads over typing	H16, M3, M36, M37, M39, M40, M41, M43, M38, L29, L31
7. Timeline edits (FIXED 9 Oct)	One write per gesture, lengths and links respected	M8, M24 to M28, M30, M32 to M35, L19 to L27
8. Local engines	Cancel, kill and load rules for LTX / H3 / Krea / audio	M18 to M23, L13, L18
9. Supplier lanes	Model and ref handling per supplier	H17, M11, M13, M15, M16, M42, M44, M45, L10, L28, L30, L32, L34, L35
10. Build	The installer contents	H18, L14
11. Standalone	One-off fixes	M1, M2, M7, M9, M31, L15, L16, L17, L33, L36

Cost of the re-audit: five Opus agents (about 3.2M tokens) plus my check.

Astra's archive: what to bring back

Compared: the rollback base (vi-dancer `5f8a25b`, koan-shared `4b46f10`) against the archive (`d7b8190`, `bedf419`). That diff is Astra's work only: vi-dancer 180 files (+14,113 / −3,917), koan-shared 68 files (+3,019 / −2,034). Each piece comes back as its own reviewed diff, one per run, without the guards.

Bring back (small, clean, matches a proven finding or the universal set)

Piece	Files	What it does
Settings wipe (fix written 8 Oct, uncommitted)	`src/main/settings.ts` → only `loadRawSettings`	Only a missing file means empty; any other read error stops the save and keeps the file. Leave its "refuse keys when the key store is off" change out.
H6 Bridge retry	`bridge/index.ts` (`took ||` one line), `bridgeAuto.ts` (`accepted` flag)	No retry on another login or supplier once a job was taken.
H18 installer leak	`package.json` (`@koan/*` moved to devDependencies), `electron.vite.config.ts` (drafts → empty in production builds)	Private source and draft MOODS stop shipping. Its `audit-package.cjs` / afterPack check is heavy (decrypts settings to scan); a simpler "no .ts/.tsx in app.asar" check does the job.
Button states	koan-shared `styles/koan.css` (+10)	secondary, danger, pressed (`aria-pressed`), focus ring, `.koan-form` field recipe
Tokens	koan-shared `src/renderer/tokens.ts` (+6)	accent / focus / warn / sans aliases, control 5 and dialog 10 radii
Keyboard Section	koan-shared `Section.tsx` (+4)	Enter/Space fold, `aria-expanded`
Menus on real buttons	koan-shared `ContextMenu.tsx`	Rows become buttons, arrow/Home/End keys, submenu back with Left. Check the face: buttons don't inherit the font by default.
Keyboard Dropdown	koan-shared `Dropdown.tsx`	Arrow keys skip disabled rows, type-ahead, Escape
Shared Modal / focus / ctl / icons	koan-shared `Modal.tsx`, `dialogFocus.ts`, `controls.ts`, `icons.tsx` (moved from vidancer-ui `ui/`)	One copy in `@koan/shared` for every app; vidancer-ui keeps re-exports
Default theme file	koan-shared `styles/theme.css`	The theme block as an importable file for other apps

Look at first (real bug fixed, but mixed with other changes)

Piece	Files	Why it needs a look
One key form	vidancer-ui `Connections.tsx`, `KeyWizard.tsx` (−95), `connections.ts`	Fixes the missing-key prompt saving a discarded key under another supplier (its `test-connection-lifecycle.cjs` reproduces it); also rewrites the prompt
Bridge login refresh	`bridge/index.ts` `renew()` single-flight	Two renders refreshing one login at once; small, but untested on his accounts

Piece	Files	Why it needs a look
Setup wizard	vidancer-ui `SetupWizard.tsx` (+94 / −37)	A capability-led wizard; a UX change, his call
Assembly buttons	vidancer-ui `Assembly.tsx`	Native buttons + project-switch checks together
Queue ownership (H9)	koan-shared `store.ts` (+423 / −221)	The cancelled-parked-row fix is inside a large store rewrite; easier to write fresh
Its tests	~60 new `scripts/test-*.cjs`	Keep only the ones that test a piece we bring back

Leave on the archive

- Every LoRA / model path change (`loraShelf.ts`, `LoraShelf.tsx`, `LoraPicker.tsx`, `ImageGenSidebar.tsx`, `loraFolders.ts`, `loraIdentity.ts`, `imagePaths.ts`, `krea2Local.ts`, `ltxLocal.ts`, `h3Local.ts`, `modelStore.ts`): this is the zone that broke Krea and LoRA.
- File-grant and boundary guards (`fileGrants.ts`, `fileScope.ts`, `boundedRead.ts`, `civitaiImages.ts`, `watch.ts`, `runtime.ts` guards, `pending.ts` path checks): hostile-file threat model.
- Log redaction (`log.ts` `redactLog`, Bridge raw reply removed from main.log): his error rows point to main.log for the full reply; redaction would hide it. His logs held no keys (8 Oct low B1).
- Toolchain upgrades (electron-builder 24 → 26, electron-vite 2 → 3, vite 5 → 6): a separate decision with its own risk.
- The fal client rework (`fal.ts`, `falImage.ts`): did not fix L8 (Image Edit variants); no gain.

Not on the archive at all: Video Editor crash recovery (Part 3 #1) and the Image Edit variant loss (L8). Those are fresh work.

Part 2. Design: one small set of universal resources

His brief: "the entire app needs to use univeral resources all buttons windows, stules, divs wrappers ... a small univbersal set of resources not hudreds fo different one, a change ot one button shoudl change all the grammer".

Measured on the restored renderer (koan-shared `vidancer-ui/src` + `src/renderer` + `styles`): 177 files, 47,116 lines, 89 React files. Script: counts every recipe, not a sample.

Where the app is today

The app is built mostly from inline styles. A shared face exists for a few things, but most surfaces hand-roll their own copy, so changing the shared one does not change them.

Building block	Shared resource today	What the app actually uses	Variants
Button	`.btn` + `.vd-hbtn` + `.ghost` in `koan.css` (30 lines)	294 elements carry `.btn`; **273 are** `<span>`, 21 are `<button>`. 130 add inline overrides (opacity as a fake disabled 38, fontSize 16, color 10, fontFamily 9). Another ~145 clickable spans/divs have no button class at all.	1 face + 130 local edits + ~145 hand-made
Icon	`Ic` (Lucide, a stand-in per ch.03/09) + 4 hand-drawn icons + the ch.03 text-glyph vocabulary	`Ic` 231 uses; ~222 text glyphs as icons. ~140 are in the ch.03 vocabulary (✕ 78, → 13, ▶ 9, ⑂ 9, ★ 7, ⚠ 9, ✓, ↔, ✂). **~80 are outside it**: + 34, ♪ 14 (ch.03 replaced ♪ with the drawn score icon), ↺ 8, ↻ 4, ⧉, ◐, ⤢, ⟲, ●○, ⎀, ✎, ⇄, ☆	vocabulary + ~80 strays
Text size	none (ch.01 scale has 7 sizes)	**19 font sizes** incl. 8, 8.5, 9.5, 10.5, 11.5, 12.5	19
Corner radius	`radius` token (4/6/8), ch.14 says 4/5/6/10	**26 radius values**	26
Padding	`spacing` token (2…16)	**105 distinct paddings**	105
Gap	same scale	11 gaps (10 and 14 outside the scale, 41 + 8 uses)	11
Colour	theme vars in `styles.css`	41 vars used well, but **51 stray hex colours** outside the theme file (Timeline 35, TimelineTransitions 15, ImageEditor 11, RenderPanel 11, `TAKE_COLORS`) + 40 rgba literals	—

Building block	Shared resource today	What the app actually uses	Variants
Layering	none (ch.01 law: one z ladder, one constants module)	**22 z-index values** from 0 to 9999	22
Field (input)	`ctl` recipe	73 of 82 inputs use it. Good.	2
Select	house `Dropdown`	50 Dropdown, **8 native** `<select>`	2
Dialog	`Modal` (ui/Modal.tsx)	13 Modal uses; ~12 other hand-made fixed overlays (DissolveModal, WatchTimelineModal, SheetGen, RenderPanel, Timeline, CoDirectorTether…)	~6 shells
Menu	`ContextMenu`	36 uses; plus hand-made popovers (workspace menu, header menus, 39 box-shadow popovers in 18 files)	several
Card / section	`Section`	19 Section uses; **111 hand-made bordered cards in 43 files**	many
Label (uppercase micro)	`.field .lb`	35 inline copies, 7 recipes (9.5/1 ×20, 9/1 ×6, 11/.5, 11/1, 10.5/.5, 10.5/2.5, 10/2)	7
Heading	none	10 size/weight combos (11/600, 11/700, 13/700, 8.5/700, 12.5/600 …)	10
Take card	`AudioTakeRow` (Score, Voice, SFX)	Clip Gen (RenderPanel), Image Gen, Sheet Gen, Image Editor tree rows, Video Editor tree nodes each draw their own. Drag targets differ by type (see Part 3).	6
Generator shell	none	10 panels each write their own render footer (now + queue + cost), 7 their own MODEL · VIA row	10
Tile (thumb grid)	none	Bin, Library, Watched, Civit, Image Gen each their own	5
Chip / badge	`LoraChip`, `ArkChip`	folder chips, tags, cost chips hand-made	several
Tooltip	native `title`	718 tooltips, unstyled, ignore the theme	1 (native)
Empty state	`EmptyState`	28 uses. Good.	1
Toast / confirm	`toast`, `confirmDialog`	192 / 37 uses; 1 `window.confirm` left	1

What that costs him

1. **One change doesn't reach everything.** Change `.btn` and 130 buttons keep their own size/colour/fade, and ~145 click targets aren't buttons at all.
2. **Keyboard and screen readers miss most of the app.** 439 clickable spans/divs, **0 with tabIndex**, 5 with a role. 33 aria-labels in the whole renderer. Tab skips them; Enter does nothing.
3. **Fake disabled.** 38 buttons fade with opacity but stay clickable; a real disabled state needs a real button.
4. **Themes leak.** 51 hard-coded colours don't follow a theme change (Timeline is the worst).
5. **Layer fights.** 22 z-index values; menus, dialogs, toasts and drag ghosts can stack in the wrong order (ch.01 already calls this a wart class).
6. **Icon strays.** ~80 glyphs carry meanings the ch.03 vocabulary never registered (law: "new meanings enter this table before they enter an app"), and 14 ♪ survive the drawn score icon.

Known since July. KOAN.design ch.09 (the vi-dancer gap checklist, 2026-07-29) already lists most of this: one context menu, one input recipe, one ✕ close idiom, one legend recipe, bare `.lb`, type-scale migration, keyboard-focusable buttons, `--warn` / `--focus`, accent var rename, tokens adoption, one z-index module, icon library migration. None is ticked. Two of its items (radius 5 → 4, mono chrome) were superseded by his 6 Oct call (ch.14: Video Dancer's sans controls and 5px buttons are the reference).

Visual pass (10-08, his running app, workspaces IMAGE GEN and KOAN MAIN)

Looked at: header, Watched, Bin, Library, Image Gen and Image Editor (empty states), Clip Gen with a clip open (takes, render footer), Clip Monitor, Timeline, Co-Director, Queue, Models, Effects. Nothing was rendered or edited; workspace and window size were put back.

1. **One face for everything.** Filled amber is the hero (▶ Render now) and also the secondary action beside it (+ Queue), the 8 "+ clip / + timeline / + char sheet …" buttons stacked down the Bin, the header (New, Project, Open, Snapshots), the Co-Director chips (room, tether, renders on, brief, compact), every Models row (update, resume, download) and the tab-like toggles. Ch.03's law is one hero per surface; the eye has no hero to find.
2. **Toggles don't show which one is on.** Watched "local | civit" and the LoRA panel's "local | civit | official" are identical filled buttons side by side.
3. **Disabled reads as a second colour.** A disabled button is the same amber at 45% opacity, which reads as a brown button (Watch cut, Cross dissolve, A−), not as "unavailable".
4. **Too small on his screen.** On the 43" 4K monitor the 9–11px type is tiny and most panels are empty space. The app has no UI scale anywhere (no `setZoomFactor` in the code). One global scale in Settings → Appearance fixes every surface at once, because the tokens scale with it.
5. **Strong already:** Clip Gen's take row (S-B card, ★ T1 with the amber outline, + edit card), beat bands under the monitors, the floating empty states, Library tiles with @titles, the Models card's ✓ status lines.

What this changes in the set: `Button` variants come first in the migration (one filled hero per surface, everything else secondary, ghost for toolbars), `Toggle` gets a drawn on-state, disabled becomes a dim neutral face with its reason, and a **UI scale** token joins the tokens.

Fix in place (no new components needed)

Fix	Size
8 native `<select>` → `Dropdown`	small
1 `window.confirm` → `confirmDialog`	one line
Hand-made fixed overlays → `Modal`	~12 sites
51 stray hex → theme vars (new extension vars where a colour has a meaning, e.g. take colours)	small–medium
z-index → one ladder module (content, dropdown, popover, modal, bootstrap, context menu, prompt, toast, drag)	small
Odd sizes → the scale (8/8.5 → 9, 9.5 → 9 or 10, 10.5 → 10 or 11, 11.5/12.5 → 12)	mechanical
`<span className="btn">` → `<button type="button">`	mechanical, 273 sites; check nested clickables and drag rows

The universal set (proposal)

One token file, one stylesheet, ~15 components, all in `@koan/shared` so KOAN.img, Sprite Dancer and shdw.work get the same set. Faces live in CSS classes, not inline styles. Inline styles keep layout only (margin, flex, width).

Tokens: colour slots (exist) · type 9 / 10 / 11 / 12 / 15 (+20 logo) · spacing 2 / 4 / 6 / 8 / 12 / 16 · radius control / field / card / dialog · z ladder · shadows · one focus ring.

The two that matter most come first, because the whole workflow runs on them: takes are the material (Part 3), and every generator makes them.

#	Resource	Replaces
T	`Take` : one take card for every list, grid and tree in the app. Layouts stay exactly as they are; only the code behind each card is shared. Same marks (★ flag, colour, disabled, ~), same keys (click, Ctrl/Shift, arrows, ~), same right-click menu, same drag-out targets, several at once. Faces per medium: filmstrip (video), picture (image), waveform (audio).	Clip Gen take row, Image Gen takes, AudioTakeRow, Sheet Gen takes, Image Editor + Video Editor tree nodes

#	Resource	Replaces
G	`Generator` **shell**: MODEL · VIA row, a recipe slot, refs, output dials, cost line, ▶ now / + queue, the takes library underneath. The dials come from each model's own capability entry (`shared/models.ts` , `imageModels.ts` , `audioModels.ts`), so video, image and audio models show only what they support and grey the rest with the reason; quirks stay in the model entry. Each panel keeps its own fields in the recipe slot (beats, characters, LoRAs, lyrics) and its own extra actions (build sheet, compose). Nothing is lost; only the duplicated frame is shared. The gen surfaces wear it whole, in the registered card split (ch.12.13: Takes / Model / Prompt / Beats). The editors (Image Editor, Video Editor) keep their tree layout, as ch.12.13 says, and take only the MODEL · VIA row and the render footer.	the frame of Clip Gen, Image Gen, Sheet Gen, Char Sheet, Enhance, Score, Voice, SFX; the model row + footer of Image Editor and Video Editor
1	`Button` (primary · secondary · ghost · danger; normal · compact; icon; pressed; disabled with reason)	294 `.btn` sites, ~145 hand-made click targets, 130 overrides
2	`IconButton` (glyph only, label required)	✕ + ★ ↺ … emoji buttons
3	`Icon` (Lucide + hand-drawn, one size scale)	~220 emoji glyphs
4	`Field` / `TextArea` (label built in)	`ctl` + `.field .lb` copies
5	`Dropdown`	native selects
6	`Toggle` (checkbox · segmented · switch)	`vd-mode` , RenderSwitch, checkboxes
7	`Slider` / drag-number	range inputs, weight chips
8	`Label` + `Heading`	7 label recipes, 10 heading combos
9	`Card` (= `Section` , foldable or not)	111 hand-made cards
10	`Row` (non-take list row: select · drag · right-click)	Library audio/video rows, LoRA rows
11	`Tile` (asset grid cell: title, badges, selection, ~ tag)	Bin, Library, Watched, Civit tiles
12	`Chip` / `Badge`	LoRA, Ark, folder, tag, cost chips
13	`Menu` (context + popover, one)	ContextMenu + hand-made popovers
14	`Modal` + confirm / prompt	fixed overlays
15	`Tooltip` (one themed layer that reads the existing `title` attributes, so 718 call sites stay as they are)	native tooltips
—	Keep as they are: `EmptyState` , `Toast` , `FloatPick` , `JogBar`	—

Enforced, not hoped for: a small check in the test suite fails the build when a face property (colour, fontSize, radius, padding, background) sits inline on a shared component, when a clickable span/div appears, or when a hex colour appears outside the theme. That keeps "change one, all follow" true after the migration.

Source of the first cut: Astra's archive branch (koan-shared `bedf419`) already holds secondary/danger/pressed/focus states in `koan.css` , a shared `Modal` , `dialogFocus` , keyboard `Dropdown` and button-based `ContextMenu` . These are leads to bring back as reviewed diffs, without its path guards.

The design guide: written as KOAN.design chapter 15, THE UNIVERSAL SET against the restored code (appended at the end of this PDF). Ch.14 (Astra's) named shared pieces that only exist on the archive branch; it now carries a dated status note saying so, and so do ch.12.15/12.16.

His calls (8 October 2026)

- **Radius: three**, 4 controls · 6 cards · 8 dialogs (ch.01's values; buttons 5 → 4, dialogs 10 → 8).
- **Tooltips: themed** (one layer reading the existing `title` attributes).

- **Glyphs stay.** "those glyphs make sense to me". All strays are now registered in ch.03 with one meaning each. ♪ → the drawn score icon (14 places). The two overlaps resolve the Premiere way: menu on-states become ✓ (track-menu solo, Window menu open panels); ◇ stays the keyframe button; ● stays the status lamp.
- **Visual pass: yes.**

Part 3. Workflows, end to end

Sources: `docs/WORKFLOW.md` (the laws), `docs/workflow-audit-2026-07-18.md` (placement law), the public AI-MANUAL (0.26.0), the panel registry (`App.tsx` PANEL_DEFS), the code paths named below, Astra's 05-WORKFLOWS as leads, and the comparison apps listed at the end.

Correction (10-08, his catch): the first draft of this part treated takes as versions of one shot to compare and pick a winner. That is wrong. It came from `docs/WORKFLOW.md` ("Keeper ★ = current best"), which predates keepers becoming flags (list 6: "a keeper — a flag, many per set", `takeRows.tsx`). WORKFLOW.md still teaches the old model; that is a doc fix for his list.

The working model. Sources (Clip Gen, Image Gen, Score, Voice, SFX, Sheet Gen, the two edit trees) produce **libraries of takes**. Variation is deliberate: he changes settings between takes, or batches 20 variations, to have many different takes to work with. Takes are the material. He marks them (★ flag, colour, disabled, ~ tag) and pulls individual takes out into use: onto the timeline as pinned blocks, into the Bin as new clips, into the Library, into ref slots, onto fields to restore a recipe, into the editors, into the next generation. There is no one right take; selecting one never retires the others. The ★ flagged last is the one dynamic blocks play.

The loop: **make variety → mark it → pull takes out into use → feed them back into the next generation.** Every stage can bounce back to the plan.

The shape today

- **24 docked panels** plus the Director's Room, the Tether, folder panels, timeline windows and Civitai examples.
- **12 generator surfaces**, each built on its own: Clip Gen, Image Gen, Image Editor, Video Editor, Sheet Gen, Char Sheet, Enhance, Score, Voice, SFX, Effects, Assembly. The render footer (render now + queue + cost) is written separately in 10 files; the MODEL · VIA row in 7.
- **6 take card implementations** (Clip Gen, Image Gen, AudioTakeRow for Score/Voice/SFX, Sheet Gen, Image Editor tree, Video Editor tree). Each take type drags to a different set of places, and every take drag carries **one** take, even when several are selected:

Take type (drag type)	Drops on	Can't drop on
Clip Gen video take (`MIME_TAKE`)	Timeline, Bin, Library, Clip Gen fields, ref slots (RefSuite), Enhance	Video Editor
Image Gen take (`MIME_GENTAKE`)	Library, Image Gen ref slots	Timeline, Bin, Clip Gen ref slots, Image Editor
Image Editor node (`MIME_EDITNODE`)	Library	Timeline, Bin, ref slots
Video Editor node (`MIME_VEDITNODE`)	Bin, Timeline	Library, ref slots (bake menu only)
Score / SFX take (`vd/score-take`)	Bin, Timeline	Library
Voice take (`vd/voice-take`)	Bin	Timeline, Library
Sheet Gen take	(not draggable)	everywhere

Library images and Bin clips already drag several at once (`MIME_IMAGES`, `MIME_CLIPS`; the timeline lays them in order). Takes don't. - **No cross-source take view.** Takes live only inside their source's card; the Bin shows a source by its keeper. Nothing lists every take in the project.

Activity by activity

Activity	How it works now	Strong	Friction (evidence)
Setup	Setup wizard, Settings → Connections (5 pages), Bridge logins, Models card	Keys tested free; Bridge auto picks the fullest account; cost on the button	Four doors to "connect something" (wizard, Settings, Bridge, Models). Bridge retry after a taken job = H6, plus H11 and M13.
Projects	New / Open grid / folders / snapshots / send to project / cleanup	Autosave, atomic saves, cloud-sync handling, send-to copies everything a timeline needs	Fine. Cleanup scan and apply are two separate passes (Part 1, low).
Gather	Library (images, music, video), Watched (local + Civitai likes), paste, browser drag, PNG reader	Every way in works; disable instead of delete; ~ tags for Co-Director	Library vs Bin vs Watched vs folder panels: four asset surfaces with different tile code (Part 2).
Plan	Clip Gen fields, beats bar, refs + @mentions, characters + links, char sheets, Style Sheet, MOODS, INTENSITY, Co-Director writes fields	Beats compile to exactly what the model sees; plan always one click away (S-B)	Identity lives in two systems (char sheets for characters; Sheet Gen plates + Library images for places/things). Panel title "Style Sheet" vs manual "Project Styles".
Generate video	Clip Gen: MODEL · VIA, setup, prompt, refs, ▶ Render now / + Queue	Capability-driven dials; refuses before spending; frozen recipes	Each generator rebuilds this same block (above).
Generate images	Image Gen (Krea local, LoRA sidebar), Image Editor (tree, select, draw, crop, compare)	Compare view, select + paste-back, external editor round trip	Two different jobs, rightly two panels: Image Gen makes batches of takes, Image Editor refines one picture in a tree. Their outputs don't travel alike: Image Gen takes and Image Editor nodes can't go to the timeline, Bin or Clip Gen ref slots by drag.
Mark and pull takes	Take cards: click select, Ctrl/Shift multi-select, ~ flag, colours, disable, double-click → Clip Monitor, drag out	One click grammar everywhere since list 2; recipe restore by drag	Drag carries one take even with several selected; drop targets differ by take type (table above); Sheet Gen takes don't drag; no way to see takes across sources or filter them by his marks.
Edit with AI	Video Editor tree (Omni edit, Kling O3 edit, Seedance re-gen), Enhance, Sheet Gen	One editor for v2v; window = only the edited piece uploads	**No crash recovery**: `vedit:run` (index.ts:1615) writes no pending entry, yet AI-MANUAL §13 says tree generations have "crash recovery". A paid edit in flight during a crash is lost.
Audio	Score Gen (compose to the cut), Voice Gen, SFX Gen; timeline lanes with keyframes and fades	Premiere-exact keyframes/fades; one math for monitor and export	Three cards, one shared take row; fine.
Cut	Timeline (Premiere mechanics), monitors, effects stack, dissolves, loops, Assembly	Deep Premiere parity; storyboard stays visible on blocks; a block pins to the exact take dragged	Panel preference bug: Assembly prefers `timelineeditor`, the panel id is `tleditor` (App.tsx:121).
Export	Timeline ⬆ Export → MP4 H.264 480/720/1080p; Bin clip → export video; Export As per item	Monitor and export share one effect chain by construction	One format, one timeline at a time, not in the Queue. Multi-timeline export agreed but unbuilt (memory pin). Preview/export parity is claimed by construction, not measured here (D3: OPEN, needs a render test).
Co-Director	Panel, Director's Room, Tether, MOODS, plans with GO, job continuation	Reads the real project, writes undoable fields, watches video, never spends without GO; plans bounded by allowances	Sees ~ tags and marks, but has no cross-source take filter to ask with (see #4).
Jobs	Queue panel: running / waiting, promote, reorder, pause, cancel; parked rows per project	Honest ETA from history; jobs stamped to their project	Cancelled parked row returns via the re-queue offer (H9). Image Gen renders never appear in the Queue (M1). Panels raise themselves on events (placement D6, still open; no calm mode exists).
Find anything	Window ▾, right-click everywhere, ? cheat sheet	Placement law (every action on right-click)	**No search.** 24 panels, hundreds of actions, no command palette.

Suggestions, ranked

Each one names the real-world pattern it borrows. None is built; each is a list item for his pick.

#	Suggestion	Why	Borrowed from
1	**Video Editor gets crash recovery** (a pending entry like `render:run`), or the manual stops claiming it	Money law; the manual currently says something the code doesn't do	the app's own `pending.ts` contract
2	**Every take travels everywhere its medium allows, several at once.** One shared take card (layouts unchanged). Image takes and image edit nodes reach the timeline (as stills), the Bin (as clips) and every ref slot; voice takes reach the timeline; Sheet Gen takes drag; editor nodes reach ref slots. A drag with several takes selected carries all of them: the timeline lays them in order, the Bin makes one clip each, ref slots fill in order.	Takes are the material; today what you can do with a take depends on which panel made it (table above), and multi-select stops at the drag	the app's own Library/Bin multi-drag (`MIME_IMAGES`, `MIME_CLIPS`)
3	**Generator shell** (Part 2 G): one frame for all generators, dials from each model's capability entry, panel-specific fields and actions in their own slots, nothing lost	One change reaches every generator; a new engine lands in one place	LTX Studio's single shot generator
4	**A project-wide take library**: every take from every source in one view (a panel or a Bin mode), filtered by his marks (★, colour, ~, disabled) and by medium, engine, source and prompt words, saved filters, drag out from there. The Co-Director reads the same filters.	Takes are scattered across their source cards; he works with them as one library	Lightroom Classic Library Filter bar (flags, colour labels, text/attribute/metadata filters); Midjourney Organize (filters, prompt-word search, saved searches, folders)
5	**Variation on the card**: each take shows what its recipe changed against its siblings (seed, prompt words, LoRA weight, model). Right-click **vary this** (re-run with one dial changed) and **sweep** (N seeds, or weights 0.6 / 0.8 / 1.0) fill the library on purpose. The Co-Director batch already does this in chat; this puts it under his hand.	Deliberate variation is how he works; the recipe difference is what makes a take useful later	Midjourney Vary (Subtle / Strong); A1111 / Forge / ComfyUI X/Y/Z plots (seed and LoRA-weight sweeps)
6	**Command palette (Ctrl+K)**: every action, panel, clip, take and setting by name, with its shortcut and where it lives	24 panels and no search; "I can't find features on my own app" (07-18)	Figma Quick Actions (Ctrl+/), Linear Ctrl+K + ? list, Blender F3 (shows the menu path)
7	**Export through the Queue**: export jobs as Queue rows, presets (H.264 / ProRes / image sequence), several timelines at once, and **export selected takes** as files in one go	One place for all long work; multi-timeline export is already agreed	Premiere → Media Encoder queue; Resolve Deliver render queue
8	**Calm mode** for auto-raising panels	Placement D6, still open	—
9	**Elements** (HIS CALL): char sheets widen to character · location · prop, one @mention grammar, Sheet Gen plates feed them	Identity for places and things currently lives as loose Library images	LTX Studio Elements (characters, objects, locations, @tag)
10	Small: Assembly panel preference id, "Style Sheet" vs "Project Styles" naming, one verb per action across generators, and `docs/WORKFLOW.md` brought up to the keeper-flag model	Same thing, same name (law 3); the stale doc misled this audit	—

Dropped from the first draft: takes as "versions" with a pick-one compare funnel; fixed meanings for his colours; Assembly "runner-up" takes. A side-by-side view of any two takes can stay as an optional tool (it helps decide which take goes where), never as a step that retires takes.

Already at or ahead of the reference apps

- Storyboard → shots → video with the plan inspectable (LTX Studio's core pitch).
- Change one moment inside a shot: the Video Editor window (LTX "Retake").
- Images in selection order → a clip with beats (Premiere 26.3 Firefly Boards → sequence).
- Premiere-grade timeline: overwrite/insert, ripple/roll/slip, keyframes, fades, J/K/L.
- Cost honesty and refuse-before-spend, which none of the reference apps show this plainly.

Open in Part 3

- D3 preview vs export parity: needs one real export compared frame by frame.
- A visual walk of each panel in his running app (Part 2 waiting list).

Sources (Part 3)

- LTX Studio storyboard + Elements: https://ltx.studio/blog/top-ltx-studio-features · https://website.ltx.studio/release-notes · https://ltx.io/blog/how-to-storyboard
- Premiere 26.3 (Firefly Boards → sequence): https://larryjordan.com/articles/new-features-in-adobe-premiere-26-3/ · https://helpx.adobe.com/sa_en/premiere/desktop/whats-new/release-notes.html
- Resolve 20 AI IntelliScript: https://www.blackmagicdesign.com/media/partial/release/20250404-02 · https://www.newsshooter.com/2025/04/04/blackmagic-design-davinci-resolve-20-announced-with-100-new-features-including-ai-enhancements/
- Lightroom Classic flags, labels, Library Filter bar: https://helpx.adobe.com/lightroom-classic/desktop/organize-photos-in-lightroom-classic/flag-label-rate-photos.html · https://helpx.adobe.com/lightroom/help/finding-photos-catalog.html
- Midjourney Organize and Vary: https://docs.midjourney.com/hc/en-us/articles/33329462451469 · https://docs.midjourney.com/docs/variations
- X/Y/Z plots (seed and LoRA-weight sweeps): https://www.stablediffusiontutorials.com/2024/09/xyz-plots.html · https://learn.rundiffusion.com/xyz-grids-comfy-auto1111
- Command palettes: https://docs.blender.org/manual/en/3.3/interface/controls/templates/operator_search.html · https://linear.app/enablement/guide/navigating-linear · https://verpex.com/blog/marketing-tips/essential-figma-shortcuts-you-need-to-know

KOAN.design · 15 — THE UNIVERSAL SET

The design guide for building any KOAN surface. Written 8 October 2026 from the restored code (koan-shared `4b46f10`+, `vidancer-ui`) and the Video Dancer audit (`vi-dancer/docs/AUDIT-2026-10-08.md`, Part 2). Chapters 00 and 01 stay the normative spine; his 6 October call (chapter 14: Video Dancer's faces are the reference) stands. This chapter says which resources exist, which are still to build, and the rules that keep them universal.

His brief: "the entire app needs to use universal resources: all buttons, windows, styles, divs, wrappers. A small universal set of resources, not hundreds of different ones. A change to one button should change all the grammar."

The rule

Every visible thing in a KOAN app is drawn by one resource from this chapter. Change the resource and every place it appears changes with it. A surface that needs something the set lacks adds it to the set first, then uses it.

Three rules that keep it true

1. **Faces live in the resource.** Colour, font, size, padding, radius, border, background and states belong to the resource (a `koan.css` class or a shared component). A call site sets layout only: margin, flex, width, gap, position.
2. **Every click target is a native** `button` (or a resource that renders one). No clickable `span` or `div`. Real `disabled`, real focus, Enter and Space for free.
3. **Values come from tokens.** Colours only through theme vars; sizes, spacing, radii and layers only through `tokens.ts`.

A check enforces all three (see "The check" below). Without it the set drifts back, which is how Video Dancer reached 2,041 inline style blocks.

Where Video Dancer stands (8 October 2026, measured)

Measure	Today	Target
Click targets that are real buttons	82 of ~520 (273 `.btn` spans, ~145 other clickable spans/divs)	all
Click targets reachable by keyboard	0 of 439 spans/divs have `tabIndex`	all
`.btn` elements with inline face overrides	130	0
Font sizes	19	the ch.01 ladder (7)
Corner radii	26	3 (4 · 6 · 8)
Paddings	105	spacing scale combinations
Hard-coded colours outside the theme file	51 hex + 40 rgba	sanctioned scrims only (ch.01)
z-index values	22	one ladder module
Take card implementations	6	1
Generator frames	10	1

Tokens

One file, `@koan/shared` `src/renderer/tokens.ts`, mirrored as CSS vars. Values follow ch.01 and his 6 October reference (ch.14).

Token	Values	Notes
Colour	the ch.01 var contract + Video extensions (`--music`, `--score`, `--sfx`, `--alink`, `--enh-*`, `--b1…--b5`)	`--warn` and `--focus` are still missing (ch.09)

Token	Values	Notes
Type	9 micro-label · 10 metadata · 11 controls and chrome · 12 inputs and menus · 13 body · 15 headings · 20 logotype	8, 8.5, 9.5, 10.5, 11.5, 12.5 snap to the nearest step
Font	`--sans` controls and prose; `--mono` code, time, cost, data	his 6 October call
Spacing	2 · 4 · 6 · 8 · 12 · 16	10 and 14 map to 8 or 12
Radius	**controls 4** (buttons, fields, chips, menu rows) · **cards 6** (cards, groups, wells, dropdown cards, toasts) · **dialogs 8** (modals, windows, pickers)	his call 8 Oct: three radii. Values = ch.01's scale; buttons 5 → 4 and dialogs 10 → 8
Border	1px resting · 2px emphasis	dashed = not yet real
Layer	content · in-panel dropdown · popover · modal · bootstrap · context menu · prompt/tour · toast · drag capture	ch.01 order; numbers live in one module
Shadow / scrim	ch.01 values	always rgba black
Focus	2px `--focus` outline, 2px offset	archive `koan.css` has it
UI scale	one app-wide scale (Settings → Appearance), applied once at the window (Electron `setZoomFactor`)	**new**. Visual pass 8 Oct: on his 43" 4K monitor the 9–11px type is tiny and the app has no scale anywhere

The set

Status: **exists** = shipped in the restored code · **archive** = written on the Astra archive branch (koan-shared `bedf419`), comes back as a reviewed diff · **new** = to build.

The two that carry the workflow

T · Take. One take card for every list, grid and tree in every app. Takes are a library: the user makes variety on purpose, marks it, and pulls individual takes out into use. The card is the same everywhere; the layouts around it stay as they are.

Part	Spec
Faces	video: hover-scrub filmstrip · image: picture · audio: waveform + the house take player
Marks	★ keeper flag (many per set; the last flagged plays on dynamic blocks) · colour edge · disabled (dimmed) · ~ tag for the Co-Director
Keys	click select · Ctrl/Shift multi-select · arrows step · ~ flag · double-click opens big
Right-click	one shared base menu (keeper, colour, disable, export as, open in, reveal, delete) + the source's own entries
Drag	one payload for every take, carrying all selected takes in order. Every take reaches every target its medium allows: timeline, Bin, Library, ref slots, fields (recipe restore), editors, another gen's refs
Hover	the frozen recipe, and what it changed against its siblings

Built from: `AudioTakeRow` (exists, the model), Clip Gen take cards, Image Gen takes, Sheet Gen takes, Image Editor and Video Editor tree nodes. Status: **new**.

G · Generator shell. One frame for every generator.

Slot	Spec
Source row	MODEL · VIA. The dials below come from the picked model's capability entry (`shared/models.ts`, `imageModels.ts`, `audioModels.ts`); a dial the model lacks stays visible, greyed, with the reason (law 7)
Recipe	the panel's own fields: Clip Gen beats and characters, Image Gen LoRAs, Score beats, Voice lines
Refs	the registered ref chips (ch.12.13)
Output dials	duration, resolution, aspect, count, seed, from the capability entry
Footer	cost on the face (law 8) · ▶ now · + queue · the panel's own extra actions (build sheet, compose). Disabled says why

Slot	Spec
Takes	the takes library of T

Gen surfaces (Clip Gen, Image Gen, Sheet Gen, Char Sheet, Enhance, Score, Voice, SFX) wear the whole shell in the registered card split (ch.12.13: Takes / Model / Prompt / Beats). Editors (Image Editor, Video Editor) keep their tree layout and take only the source row and the footer. Status: **new**.

Controls

Resource	Job	Status
`Button`	every action. **One filled primary per surface** (the hero, ch.03); everything else secondary; ghost for toolbars; danger for destroy. Visual pass 8 Oct: today the hero, its neighbours, section adds, header, chips and toggles all wear the same filled amber, and disabled is that amber at 45% (reads as brown). Disabled becomes a dim neutral face with its reason. Variants: primary (filled accent), secondary, ghost, danger. Sizes: normal, compact (`.vd-hbtn`). States: pressed (`aria-pressed`), disabled with reason, focus ring. Lowercase law (ch.03). The hero morphs on its own face (ch.03)	face exists (`koan.css`); secondary, danger, pressed, focus **archive**
`IconButton`	a glyph-only action; label required for keyboard and tooltip	**new**
`Icon`	`Ic` (Lucide stand-in, 13px, 1.75 stroke) + the hand-drawn set + the ch.03 text-glyph vocabulary. A new meaning enters the ch.03 table before it enters an app. Menu rule (8 Oct): in a menu, "on" is a ✓; diamonds are keyframes; dots are status lamps	exists; strays registered in ch.03 on 8 Oct; to fix: 14 ♪ → the drawn score icon, solo ◆/◇ → ✓, Window menu ● → ✓
`Field` / `TextArea`	text entry with its label built in	`ctl` recipe exists (73 of 82 inputs); `.koan-form` **archive**
`Dropdown`	every one-of-N list; stable width law	exists; 8 native selects left; keyboard **archive**
`Toggle`	binary = checkbox; one-of-N small = segmented (`.vd-mode`); render on/off = switch (ch.03 one job, one widget). The on-state is drawn (law 1): visual pass 8 Oct found "local / civit / official" as three identical filled buttons	parts exist, **new** as one resource
`Slider`	a value with its live readout, always paired (ch.03)	**new** (10 range inputs + weight chips)
`Chip` / `Badge`	ref chips, LoRA chips, folder chips, stat chips (SPND, FAL)	`LoraChip` , `ArkChip` exist; rest **new**
`Jogger`	the media transport bar (ch.03 canon)	exists

Structure

Resource	Job	Status
`Label`	the 9px uppercase tracked micro-label (ch.04)	**new** (35 inline copies, 7 recipes)
`Heading`	panel and card headings, legends (ch.09 "one legend recipe")	**new** (10 combos)
`Card`	the bordered group / Section, foldable or not (ch.04)	`Section` exists (19 uses); 111 hand-made cards; keyboard fold **archive**
`Row`	non-take list rows: Library audio/video, LoRA rows	**new**
`Tile`	asset grid cells: Bin, Library, Watched, Civit (ch.04 tile)	**new**
`Panel`	header / body / footer anatomy (ch.04)	recipe exists in prose; **new** as a wrapper

Overlays and feedback

Resource	Job	Status
`Menu`	right-click and popover menus, one implementation (law 11)	`ContextMenu` exists (36 uses); real buttons + keys **archive**; hand-made popovers to retire
`Modal`	every dialog card; one backdrop, one close idiom	exists in `vidancer-ui/ui/Modal.tsx` (13 uses); shared copy **archive**; ~12 hand-made overlays to retire
Confirm / prompt / alert	questions and answers (ch.05)	exists (`prompt.tsx`); 1 `window.confirm` left
`Tooltip`	the teaching layer (law 13). His call 8 Oct: **themed**. One tooltip layer reads the existing `title` attributes (718 call sites stay as they are), draws in the theme, and keeps the ch.02 voice: lowercase, gesture + consequence + shortcut, one thought per line	native `title` today; **new**
`Toast`	feedback, one surface	exists
`EmptyState`	panel-level empties (law 9)	exists
`Progress`	honest bars, indeterminate bar, spinner, the pile (ch.05)	parts exist
`FloatPick`, `PanelErrorBoundary`, `ResizeHandle`	as registered (ch.12.13, ch.05)	exist

The check

A script beside `koan-shared/scripts/drift.mjs` scans each consumer's renderer and counts: inline face properties on resource elements, clickable spans and divs, hex colours outside the theme files, z-index literals, font sizes off the ladder, radii off the tokens, native `select` and `window.confirm`. It starts as a **ratchet**: the counts in "Where Video Dancer stands" are the baseline, and the check fails the build when any count goes up. Each migration step lowers the baseline. At zero the rule is total.

Migration order

One step per run, committed, tested by him before the next.

1. Tokens: `--warn`, `--focus`, the z ladder module, the archive token aliases, the UI scale. The check, as a ratchet.
2. `Button`: archive states into `koan.css`; `span.btn` → `button` (mechanical, 273 sites; nested clickables and drag rows checked one by one).
3. `Take`: one card, one drag payload, every target per medium.
4. `Generator` shell: gen surfaces first, then the editors' source row and footer.
5. Leftovers of law 11: 8 selects → `Dropdown`, hand-made overlays → `Modal`, popovers → `Menu`.
6. `Label`, `Heading`, `Card`, `Row`, `Tile`, `Chip`.
7. Type, radius and colour snaps (mechanical once the resources own the faces).
8. Icons: 14 ♪ → the drawn score icon; track-menu solo ◆/◇ → a ✓ row; Window menu ● → ✓. The themed tooltip layer.

Nothing in this order changes what a take, a render or a timeline does. Where a step would change behaviour, it stops and asks.

How ch.09 maps here

ch.09 item	Resource / token
theme the dock chrome	Tokens (`--dv-*` overrides)

ch.09 item	Resource / token
hover states on menu rows · one context menu	`Menu`
one input recipe	`Field`
one ✕ close idiom	`Modal` , `IconButton`
one legend recipe · bare `.lb`	`Heading` , `Label`
type-scale migration	Tokens + step 7
keyboard-focusable buttons	`Button`
`--warn` , `--focus` · accent rename · raised fold · tokens adoption · one z module	Tokens
icon library migration	`Icon`
radius 5 → 4 · mono chrome	superseded by his 6 October call (ch.14)

Acceptance

Chapter 14's seven interaction checks apply to every resource: tab reaches every action and Enter/Space activates once; Escape closes the innermost layer and focus returns; menus skip separators and disabled rows; narrow widths wrap and keep the main action; every state (empty, ready, loading, cancelled, failed, success) is drawn; a custom theme keeps text, focus and disabled reasons legible; a paid job confirms once and never retries itself after a receipt.

Decisions (his, 8 October 2026)

- **Radius: three.** 4 controls · 6 cards · 8 dialogs (ch.01's values).
- **Tooltips: themed.**
- **Glyphs: they stay.** The strays were sensible; they are registered in the ch.03 table with one meaning each. ♪ becomes the drawn score icon. The two overlaps resolve the Premiere way: menu on-states are ✓, ◇ stays the keyframe button, ● stays the status lamp.
- **Visual pass: done 8 Oct** (IMAGE GEN and KOAN MAIN workspaces). Findings are in the audit, Part 2: one face for everything, toggles without a drawn on-state, disabled reading as brown, and no UI scale on his 43" 4K screen.